# The Evolution of Ransomware in Today's Digital Landscape

Kerry W. Runge, Dustin J. McKinney, Jacob L. McClain

***Abstract*--The threat of ransomware grows every year as more companies' assets are seized, held for ransom, and sometimes distributed by attackers. Ransomware is taking the world by storm attacking many high-profile actors. The diversity of ransomware attacks causes a constant evolution of tactics for detecting, preventing, and mitigating ransomware attacks. Understanding the current ransomware landscape begins with the study of the original ransomware attack, the AIDS Trojan in 1989. Attacks have changed from a one-off, high effort specialized attack to ransomware as a service that can be purchased and used, shielding the original creators from the spotlight whilst receiving a cut of the ransoms. With the rise of RaaS, smaller companies and individual users run a greater risk of being targeted and exploited. Large scale attacks are more prevalent from large criminal organizations or nation Individual users are more likely to experience a ransomware attack which can have devastating consequences, both financially and in the sense of data security. In the past individual users were rarely affected and were more likely to be affected as part of a large-scale attack against an organization. More focused attacks allow for attackers to use social engineering tactics such as spear-phishing to tailor the attack to the target. RaaS allows for more flexibility for the attacker as they do not have to program the attack but instead just execute an attack. RaaS is on the rise and is forecasted to be the future of ransomware. In this paper we will examine how ransomware is defined both in its construction and in its application, we will also explore what is happening on the defense side and how enterprises can prepare.***

## I. Introduction

Ransomware is a form of malware whose purpose is to change the authorized users' access to their data and only allow access back through a ransom. Ransomware can be broken into two distinct categories based on how it denies the user's information [7]. The first is crypto ransomware, this malware encrypts all the user's data except for what the malware needs to function or multiply across a network. Second is locker ransomware, a type that locks the user's information up front similar to a vault. In both cases the user must pay a ransom demand to receive their information back however the integrity of the data has already been compromised and the attacker can remain in the system causing more future harm. The first widespread implemented ransomware was the AIDS Trojan in 1989 [10]. With this variant a user's files were encrypted, and a ransom was demanded for the return of the files. As the date may suggest this type of malware has been around for a long time, however there was no secure way to transfer the payment to the attacker. This attack was also unique in that the way it spread was through floppy disks that were handed to individuals during an AIDS conference. This made attacks of this kind rare due to their payments being traceable. Peer to peer cryptocurrencies fixed this problem for threat actors because of the anonymity of the ransom demands. This change in how payments can be exchanged gave way to one of the worst ransomware attacks. In 2017, a ransomware called WannaCry shocked the world hitting any industry it could touch. If the device was connected to the internet, it was threatened by WannaCry due to the MS17-010 "EternalBlue" SMB vulnerability [10]. The program locked the user out of their data and left two files unencrypted. The first file instructions for the user to send the ransom demand via Bitcoin and the other file was the decryption program. WannaCry overwrites the contents of files by opening the file, reading the file, encrypting the file, then closing the file. The scope of WannaCry was immense and Madhya Pradesh, a region in India, was hit the hardest with 32.63% of attacks being WannaCry. In the United Kingdom, the NHS had at least 16 organizations completely shut down crippling their healthcare industry. In China, universities were hit, with students' and professors' work being encrypted [10]. WannaCry launched a new age of ransomware attacks bringing with it an exponential growth of new malware programs.

## II. Taxonomy of Current Ransomware Families

### A. Weaponization

Despite new malware programs being created the general structure of the program remains the same. The variation between the malware programs comes in how they are delivered and what encryption algorithm is used. Each ransomware program contains a weaponization method, evasion method, delivery method, exploitation method, installation method, and a Command and Control (C2). There are several variations on how malware can be weaponized, the first is through a script, weaponized macro, or malicious executable that has the executing commands. Once the commands have been run the original script file is deleted and the opcodes are written to memory. Malware that implements this is called file-less or in-memory-only malware [6]. This is a stealthier way to install ransomware and is more common for large scale attacks and can be used to bypass host-level disk scanning anti-malware to infect limited privileged users. Most script-based malware uses JavaScript, PHP, PowerShell, and Python to create command scripts. Sending a simple executable file is more difficult than it sounds because anti-malware

products can block the download or execution of executable files. Malware can use normal looking files to hide the malware executable files to bypass anti-malware programs and to convince users to run the program. Common file types used are: SVG, DOCX, ZIP, HTA, SFX, VBS, DLL, and PDF. Even if a ransomware file was executed anti-malware programs can still catch the encryption process and kill it. Ransomware programs diversify how they access and encrypt the host files [6]. For example, a prominent ransomware family, *Locky*, will change file extensions to .locky to evade detection. Other file extensions seen are .crypt, .exx, .ezz, .exx, .xyz, or .zzz. Other ways of bypassing detection are by overwriting the encrypted data on the same data buffer or by deleting the original files by removing entries from the Master File Table [6]. Ransomware malware all share the common goal of encrypting information and demanding a ransom. Newer ransomwares have switched to exfiltrating data and demanding the posting of sensitive information. However, different ransomware families use different encryption methods. The three categories of encryption methods using OS features or API are: Asymmetric Encryption, Symmetric Encryption, and a Hybrid Technique. The asymmetric encryption method uses a public/private key to encrypt the user's data [6]. In the case of WannaCry the keys are stored on the victim's machine which is what led to stopping WannaCry [10]. In the case of Locky, the keys are delivered through C2 channels. TeslaCrypt uses a key embedded in the binary executable. Symmetric encryption usually has a key embedded in the malware. However different families also use different patterns in their encryption. UIWIX begins by encrypting with AES-256 in CBC then encrypts again using RC4. Bucbi uses GOST which is a 256-bit cryptographic hash function developed by the KGB and declassified in 1994 [6]. The final method is hybrid. Hybrid uses both asymmetric and symmetric to encrypt the user's information. The malware begins by encrypting the user's files with symmetric key algorithms such as AES-256 and CBC followed by using asymmetric encryption such as RSA-1024, RSA-2048, or ECC to encrypt the symmetric key [6]. However, unlike asymmetric where there is C2 to retrieve the encryption key, the RSA public key is embedded in the binary payload. There are some ransomware families that use unique techniques to encrypt information such as Mischa that uses a randomly generated key as a seed to XOR the user's data for encryption.

### *B. Evasion*

Evasion is an important aspect of malware because it allows for the execution of malicious code and the creation of the desired outcome. Ransomware has four distinct categories of evasion techniques: Timing-based evasion, Data evasion, Code evasion, and Network evasion. Combining these techniques allows for ransomware to become destructive. Timing-based evasion relies on running the program at a specific time/date. Ransomware can have a delayed execution of a few minutes to days before the encryption begins. This method is adopted to allow for the program to be executed but then it can go unnoticed because it does not immediately begin encrypting the data. KeRanger, Reveton, CryptXXX, and Cerber are some of the families that implement this system to bypass scans [6]. Ransomware can also remain dormant until an event causes a trigger. This event is generally when the program finds the most vulnerable moment such as the system being idle or an event like a system reboot. KeRanger ransomware uses the event of an OSX Time Machine backup to trigger its encryption. Using this event allows the KeRanger to not only encrypt data that is yet to be backed up but also gain access to Time Machine to encrypt the user's backup.

Not only does the program need to evade anti-malware systems but also evade researchers who look to improve their detection systems. Data evasion is important for avoiding being studied. Some ransomwares will come equipped with self-deleting/self-destruction instructions. The purpose of these is to delete traces of themselves from the target machine. Ransomware can also have anti-dump measures in place. When ransomware is executed, its opcodes are decrypted and unpacked in memory so the opcodes can be run by the CPU [6]. Ransomware will utilize an anti-dump to modify/erase the PE header or implement a technique called stolen bytes in which distinct parts of binary are loaded into distinct parts of memory. If these techniques are not implemented, forensic analysts could extract the encryption keys and use them for future decryption. Ransomware can also manipulate an operating system into thinking an executable file is a legitimate file. This is done through creating an alternate data stream or ADS. The ADS allows the operating system to know the attributes of a file. Some ransomware families such as TeslaCrypt manipulate the ADS to bypass detection systems to deliver its payload. The last data evasion technique also involves manipulating the ADS by deleting the Zone Identifier extension [6]. The zone identifier shows the origins of a file, for example Windows will add a zone identifier to files downloaded from the internet through a web browser. Attackers will delete this information to hide the origin of the ransomware's files to further complicate the forensic process.

Ransomwares such as TeslaCrypt, CTB-Locker, Locky, CyptoWall, and TorrentLocker have code that detects if a debugger is active, and this will prompt the binary payload to delay. In the case of disassembling the code, ransomware developers will load junk/dead code to extend the analysis process of their programs. In an attempt to make their code unreadable, ransomware developers will add code sequences and branches that contain no execution effects resulting in more wasted time. Ransomware can also detect if it is in an isolated environment such as a VM or sandbox. This is a popular way to study malware since it cannot break out of its container and infect other machines. Many ransomware families can detect this and terminate itself or remain dormant. The program will check the TSC register to reveal the CPU cycles. In a virtual environment the cycles will differ from a live environment. Other techniques include checking the MAC address of the host machine or checking the running processes, all of which would differ from a live environment. Ransomware developers can also program their ransomware to undergo polymorphism and metamorphism. The ransomware code makes slight changes to itself to constantly attempt to avoid detection and study. The polymorphic attribute is done by altering its static binary code using a different encryption key.

The final method of evasion is evading network detection. Just how encrypting a user's information blocks the user from seeing the data. The malware will encrypt its network traffic to block detectors from detecting its communication with C&C.

Ransomware can also use traffic anonymizers to encrypt the communication between two endpoints. From the two endpoints the encrypted communication is relayed forward through several relay nodes. The ransomware family CryptoWall v3 uses I2P network proxies to communicate with C2 but uses Tor network for ransom payments [6]. CryptoWall v3 also utilizes another technique called domain shadowing to communicate. The developers steal domain credentials of legitimate domains to create numerous sub-domains. These sub-domains are then mapped to the ransomware developer's server. The evasion technique is to rotate the sub-domains obscuring the legitimate path to the developer's server, this can be done using a domain generation algorithm in an attempt to create rendezvous points on random domains. The last network evasion method is called Fast Flux. It is comparable to the rotating sub-domains except the IP address associated with a domain/DNS is mapped with other IP addresses and then they are rotated [6]. This method protects against blacklisting the IP. WannaCry ransomware implemented this technique to protect itself.

### *C. Delivery*

Despite the numerous ways in which a ransomware developer can evade detection and fool forensic researchers, malware must be delivered to the targeted user. Attackers can use the following techniques to deliver ransomware, Social Engineering, Malvertisement, and TDS. Social engineering is a common way to deliver malware because it involves manipulating the targeted user's trust to get them to perform an action for the attacker. The two common forms of social engineering are phishing and spear-phishing. Phishing is an attempt to get the user to share sensitive information usually through spam emails, instant messages, or phone calls. Spear-phishing is similar however the attacker performs reconnaissance of the target before by gathering information about them. Since spear-phishing takes more time to set up, it is used more against organizations or a specific group of people that are critical such as VP's and C-level executives. Once the targeted user has been hooked the attackers can remotely drop the ransomware payload onto the target's machine. A more passive way to trick users is through Malvertisement. Malvertisement are advertisements that redirect the user to the attacker's domain once they have clicked on the fake advertisement. Malvertisement campaigns can happen on trusted websites such as New York Times and BBC [6]. Like phishing attempts, once the user has been redirected the ransomware can be installed. Cerber ransomware family ran a malvertisement campaign that redirects users to Manitude exploit kits. Traffic distribution systems or TDS is used to redirect users from a legitimate site to a malicious site which contains the attacker's malware. The attacker buys redirected traffic from a TDS vendor to initiate the redirect and once the user is on the malicious page an exploit kit is purposed however browser-based attacks have been decreasing due to an increase in browser security.

### *D. Exploitation and Installation*

Once the ransomware has been delivered the exploitation of the system begins. Attackers can implement an exploit kit to scan for vulnerabilities in a system. When a vulnerability has been identified it uses it as a foothold to attach itself to. From the foothold the ransomware can launch its payloads. The usage of an exploit kit allows for a quicker way to identify a system's vulnerabilities or weak configurations. However, the more processes that are run the greater chance the ransomware is caught. Attackers can run reconnaissance on targeted systems to determine vulnerabilities. This style is more common with ransomware attacks targeting enterprises. When the foothold has been established the installation process begins. The ransomware deploys in two phases that can occur concurrently. The first is infecting the host machine and the second phase is infecting the network the host machine resides on. The malicious payload is launched on the host machine and seeks to encrypt all local files and then installs itself on backup versions within its reach. Not every ransomware targets the same files. Some may encrypt the user's data and deny access whereas others may target specific file types such as images, videos, PDF files, etc. Some ransomware families, such as HDDCryptor, will take a more drastic approach and encrypt the entire drive of the victim. These types are known as "boot lockers." During the encryption process the ransomware will attempt to find any other external storage devices such as USB, external HDD, or cloud storage devices to encrypt. This is an attempt to maximize the effectiveness of the program by reducing the chances of a victim's recovery. The second phase of installation is infecting the network. Ransomwares in the past have utilized a vulnerability, MS15-067 and MS15-030, in RDP to do remote code execution through the remote desktop. SMB protocol is a client-server protocol used by Windows OS to share files, printers, and serial ports [6]. Malware can use this protocol to spread itself over the shared files, devices, and ports. Two prominent SMB protocol vulnerabilities are "EternalBlue" and "EternalRomance", ransomware families such as WannaCry and NotPeyta actively use these exploits to infect the network.

### *E. Command and Control (C2)*

C2 refers to the malware's ability to communicate with the attacker's server. Traditionally malware can receive more instructions, however ransomware is unique in that C2 is used for receiving the encryption key and used for receiving ransom payments. If a ransomware family uses asymmetric encryption C2 is particularly important because the public and private key pairs must be transferred. If the keys are not transferred, then the encryption cannot occur. There are 3 unique techniques for C2, Hard coding C2 IP Addresses, Domain generation algorithms, and pre-existing botnets. The C2 IP addresses can be hard coded into the ransomware file. Locky leverages this technique because it creates less network traffic since it does not perform domain searches or DNS queries. The ransomware may be stealthier, but it is more susceptible to being reverse engineered and having the IP address blocked on the network gateway. Locky will also use domain generation algorithms to change the domain names to make tracing difficult for security teams. Lastly botnets are used to send payments and malicious code through its network. Botnets are a system of compromised machines or bots that communicate with one another through the C2 server. C2 techniques are implemented to avoid detection and allow for the

transferring of keys and payments to and from the target machine. This also allows for manual recon, exfiltration of sensitive data, and for the deployment of stagers and other malware files.

TABLE 1
Conti C2 Indicators of Compromise

| Domains | | | | |
|---|---|---|---|---|
| badiwaw[.]com | fipoleb[.]com | kipitep[.]com | pihafi[.]com | |
| balacif[.]com | fofudir[.]com | kirute[.]com | pilagop[.]com | tiyuzub[.]com |
| barovur[.]com | fulujam[.]com | kogasiv[.]com | pipipub[.]com | tubaho[.]com |
| basisem[.]com | ganobaz[.]com | kozoheh[.]com | pofifa[.]com | vafici[.]com |
| bimafu[.]com | gerepa[.]com | kuxizi[.]com | radezig[.]com | vegubu[.]com |
| bujoke[.]com | gucunug[.]com | kuyeguh[.]com | raferif[.]com | vigave[.]com |
| buloxo[.]com | guvafe[.]com | lipozi[.]com | ragojel[.]com | vipeced[.]com |
| bumoyez[.]com | hakakor[.]com | lujecuk[.]com | rexagi[.]com | vizosi[.]com |
| bupula[.]com | hejalij[.]com | masaxoc[.]com | rimurik[.]com | vojefe[.]com |
| cajeti[.]com | hepide[.]com | mebonux[.]com | rinutov[.]com | vonavu[.]com |
| cilomum[.]com | hesovaw[.]com | mihojip[.]com | rusoti[.]com | wezeriw[.]com |
| codasal[.]com | hewecas[.]com | modasum[.]com | sazoya[.]com | wideri[.]com |
| comecal[.]com | hidusi[.]com | moduwoj[.]com | sidevot[.]com | wudepen[.]com |
| dawasab[.]com | hireja[.]com | movufa[.]com | solobiv[.]com | wuluxo[.]com |
| derotin[.]com | hoguyum[.]com | nagahox[.]com | sufebul[.]com | wuvehus[.]com |
| dihata[.]com | jecubat[.]com | nawusem[.]com | suhuhow[.]com | wuvici[.]com |
| dirupun[.]com | jegufe[.]com | nerapo[.]com | sujaxa[.]com | wuvidi[.]com |
| dohigu[.]com | joxinu[.]com | newiro[.]com | tafobi[.]com | xegogiv[.]com |
| dubacaj[.]com | kelowuh[.]com | paxobuy[.]com | tepiwo[.]com | xekezix[.]com |
| fecotis[.]com | kidukes[.]com | pazovet[.]com | tifiru[.]com | |

## III. CURRENT RANSOMWARE LANDSCAPE

### *A. RaaS*

Ransomware as a Service or RaaS is a developing industry within the dark web. The dark web refers to hidden networks on the Internet such as TOR or I2P [9]. Within TOR and I2P there resides a collection of websites that can be used as a marketplace. RaaS is being sold as a franchise model to individuals on dark net forum marketplaces. Traditionally ransomware attacks are executed by a single threat actor or a threat organization. Threat actors have begun migrating towards using ransomware as a product/service rather than a tool. Developing effective ransomware products is time consuming and difficult. The development process requires a lot of knowledge regarding multiple aspects of a computer, programming, networking, operating systems, anti-malware evasion. Developing a dangerous tool and executing an attack brings a lot of attention from security agencies. Large ransomware attacks occur against targets called Big Game. Big Game refers to large enterprises, infrastructure, government sites, military systems, etc. RaaS avoids the attention of Big Game hunt and instead offers these devastating tools to smaller threat actors who wish to extort money or accomplish other goals. Implementing an attack requires little to no programming skill which significantly widens the scope of potential attackers. The ransomware developers receive a portion, 20-30%, of the profits generated by the actual attackers [9]. The developers also have the added benefit of not being in the spotlight of security agencies, but their product and consumer is. An analysis of the current dark web marketplace shows that 0.73% of transactions are RaaS offered goods.

### *B. Analysis of Current Reported Ransomware Attacks*

With the rise of RaaS and the growth of ransomware families, Cybersecurity companies are on the forefront of developing innovative technologies to thwart attackers. In reports published by SOPHOS and CrowdStrike, two security industry leaders, ransomware is cited as a major focus in their defense systems. Crowdstrike reported a staggering 82% increase in ransomware related data leaks in 2021 [3]. SOPHOS cites a pivot to RaaS model for ransomware attacks specifying Conti Ransomware. WIZARD SPIDER is the force behind the Conti epidemic in 2021 however CrowdStrike's Falcon OverWatch has been monitoring WIZARD SPIDER's ransomware. They have observed its multiple server access through its utilization of RDP to authenticate into Windows Domain Controller. Tasks are scheduled to execute in ShellStarter then Conti moves laterally to gain access to another domain controller leveraging RDP. Rogue DLL files are pushed before the execution of the Microsoft signed binary Rundll32. Rundll32.exe loads the malicious .dlls which can interact directly with the windows api to perform malicious functions. This sets up the C2 and then it can move laterally for the third time gaining access to another domain controller through Windows administrative share and set the backdoor, AnchorDNS. At this point the program was caught attempting to set up a fourth domain controller using a new administrator account [1]. The victim was able to regain control of the system and expel the threat. Secret documents were leaked by a Conti affiliate showing some operations. In their 2022 threat report SOPHOS labs breaks down Conti's ransomware tools and how they are used to manipulate initial access, execution, privilege escalation, defense evasion, credential access, discovery, lateral movements, and its impact [1]. The scope of the impacts is felt across all industries. The primary industries attacked in 2021 occurred in the industrial and engineering sectors. The least impacted sectors are logistics and academics. The year 2021 was a dangerous year ranging from malicious instructions implemented with the SolarWinds hack, to REvil ransomware attack on Kaseya, to ransomware deliveries through Microsoft's Exchange mail patch vulnerability. In October of 2020, the CISA released a joint Cybersecurity advisory with the FBI and the HHS. In this advisory it is cited that there is an increased and imminent threat to US hospitals and healthcare providers. The report cites the creators of BazarLoader, a well-known malware program, are releasing the ransomware TrickBot specifically targeting the HPH sector and engaging in data theft and disruption of operations [2]. The ransomware family used in healthcare attacks is Ryuk which was originally created from Hermes 2.1 ransomware in 2018. Ryuk uses commercial products such as Cobalt Strike and PowerShell Empire to deploy ransomware. Companies are facing a massive growing ransomware force and Cybersecurity companies are tasked with constantly adapting their systems to combat threat actors.

## IV. PREVENTION AND RECOVERY

Ransomware is very destructive and disruptive malware that can cause hundreds to millions of dollars in damages. Simple preventative measures can be put into place to avoid the disruptive nature of malware. Training employees in spotting phishing attempts and suspicious links is a fundamental practice preventing a malware attack. Employees should be trained not to click on unconfirmed links and not to open untrusted email attachments. Employees should also avoid downloading from untrusted websites and avoid giving out personal information. Administrators should keep systems up to date with the latest patches, especially security patches. Security software should be implemented, and teams should be up to date with the latest risks such as a zero-day exploits. Based on the current environment and business, suitable policies should be created and implemented. Backups should be completed and securely

stored in case of an attack. If a machine becomes infected, it needs to be separated, reported, identified, and then from there the best option for remediation should be determined [7]. Once the issue has been remedied the machine can be restored and refreshed. If the attack is widespread then the network operations should be shutdown. If the network is properly segmented, then only the infected segment must be shutdown. Firewalls should also be properly configured and a "gold image" of system configurations, in case of an attack the system can be restored [7]. Creating security guidelines and following them are important for mitigating the risk of a ransomware attack. Ultimately, even with these prevention mechanisms in place, a sufficiently motivated and funded threat actor could still infect a target organization with ransomware and demand a ransom. Even worse, they could demand any number of things while holding sensitive data hostage. Each organization should not only have these prevention and remediation mechanisms in place, but also work on an active basis to test their detection and prevention, conduct field testing of recovery plans, and make a best effort to regularly pen test their own systems.

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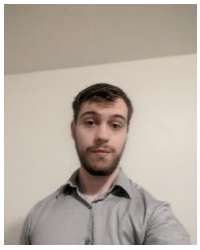

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